PG No. 43 $C_6(c)$ 6 [hexagonal]

Table 1: Representation matrices

Irrep.						
A	$1:\left[1\right]$	$3^+_{001}:\left[1\right]$	$3^-_{001}:\left[1\right]$	$2_{001}:\left[1\right]$	$6^-_{001}:\left[1\right]$	$6^+_{001}:\left[1\right]$
B	$1:\left[1\right]$	$3^+_{001}:\left[1\right]$	$3^-_{001}:\left[1\right]$	$2_{001}:\left[-1\right]$	$6^-_{001}:\left[-1\right]$	$6^+_{001}:\left[-1\right]$
$E_1^{(a)}$	$1:\left[1\right]$	$3^+_{001}:\left[-\frac{1}{2}-\frac{\sqrt{3}i}{2}\right]$	$3^-_{001}:\left[-\frac{1}{2}+\frac{\sqrt{3}i}{2}\right]$	$2_{001}:\left[-1\right]$	$6^-_{001}:\left[\frac{1}{2}+\frac{\sqrt{3}i}{2}\right]$	$6^+_{001}:\left[\frac{1}{2}-\frac{\sqrt{3}i}{2}\right]$
$E_1^{(b)}$	$1:\left[1\right]$	$3^+_{001}:\left[-\frac{1}{2}+\frac{\sqrt{3}i}{2}\right]$	$3^-_{001}:\left[-\frac{1}{2}-\frac{\sqrt{3}i}{2}\right]$	$2_{001}:\left[-1\right]$	$6^-_{001}:\left[\frac{1}{2}-\frac{\sqrt{3}i}{2}\right]$	$6^+_{001}:\left[\frac{1}{2}+\frac{\sqrt{3}i}{2}\right]$
$E_2^{(a)}$	$1:\left[1\right]$	$3^+_{001}:\left[-\frac{1}{2}-\frac{\sqrt{3}i}{2}\right]$	$3^-_{001}:\left[-\frac{1}{2}+\frac{\sqrt{3}i}{2}\right]$	$2_{001}:\left[1\right]$	$6^-_{001}:\left[-\frac{1}{2}-\frac{\sqrt{3}i}{2}\right]$	$6^+_{001}:\left[-\frac{1}{2}+\frac{\sqrt{3}i}{2}\right]$
$E_2^{(b)}$	$1:\left[1\right]$	$3^+_{001}:\left[-\frac{1}{2}+\frac{\sqrt{3}i}{2}\right]$	$3^-_{001}:\left[-\frac{1}{2}-\frac{\sqrt{3}i}{2}\right]$	$2_{001}:\left[1\right]$	$6^-_{001}:\left[-\frac{1}{2}+\frac{\sqrt{3}i}{2}\right]$	$6^+_{001}:\left[-\frac{1}{2}-\frac{\sqrt{3}i}{2}\right]$